

WP3.3:

Title: “Development of use cases’ value chains cognitive DTs”
Leader: IND



INNOVATIVE MODELLING & ASSESSMENT CAPABILITIES
THROUGH MAAS FOR MANUFACTURING ECOSYSTEM RESILIENCY

(X th December).(Online)



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-  **Overview & Objectives of Task 3.3**
-  **Current Status**
-  **Completed Work**
-  **Remaining Tasks & Next Steps**
-  **Coordination Plan with Other
Partners**
-  **Open Discussion**

Overview & Objectives of Task 3.3

[T3.3]:  ->M4  -> D3.2 

Task	Description	Leader	Year 1												Year 2												Year 3												Year 4												
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	
WP3	Manufacturing Ecosystem double level cognitive digital twins	KIT																																																	
T3.1	Characterisation of manufacturing services and use cases' value chains as cognitive DTs	KIT																																																	
T3.2	Development and validation of the manufacturing services cognitive DTs	KIT																																																	
T3.3	Development of use cases' value chains cognitive DTs	IND																																																	
T3.4	Manufacturing Ecosystem cognitive DT integration, validation and tune-up	KIT																																																	

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Current Status

Task / Methodology Item	Status	Explanation
1. Digital replication of use cases' value chains key elements	🔄 On going (Preliminary simulation model)	Core VC6 ontology developed; six-layer value chain model (Suppliers → Customers) fully defined. Preliminary DE S simulation replicates stock, production, and shipment behaviour synthetically.
2. Interconnections and dependencies between value chain stages (based on T3.1)	⌚ Not started (planned for next phase)	Forward and backward dependencies not implemented yet. Inter-stage constraints (supplier delays, warehouse congestion, transport dependencies) will be added after receiving WP2's modelling inputs.
3. Develop value chain DT for use cases	🔄 In early stage	Simple
4. Detailed description of production stages (T3.2)	🔄 On going	This component will be integrated into the simulation during the next development stages as the work progresses.
5. Cognitive modelling techniques to generate scenarios	🔴 Not yet	No AI-based scenario generation or learning algorithms implemented yet. Still under planning.
6. Validation scenarios by use cases and comparison to with real data	🔴 Not yet	Demand data was generated using KDE based on distributions. However, full scenario validation using real data is still pending.

Current Status

Task / Methodology Item	Status	Explanation
Value Chain flow structures defined for both use cases	✅ Done	High-level value chain flows have been defined for both KAM and JPB, using a common six-layer structure
• Initial simulation prototypes created	✅ Done	A first set of discrete-event simulation prototypes has been implemented with syntetic data to test the VC-DT concept.
• KPI categories for both use cases analysed	❑ Needs validation from partners	KPI families were extracted from the use case documents and mapped to the High-Level Value Chain DT (lead-time, OTD/OTIF, inventory, flow stability, transport delays, etc.) and to the Manufacturing-Level DT (OEE, FPY, downtime, scrap). This classification now needs partner validation to confirm priorities and expectations.
• Data requirements table prepared	✅ Done	An input–output data requirements table has been prepared for the six value-chain layers.
• Received data mapping file from partners	❑ Needs validation from partners	The file shows where key fields are stored; for now we just need confirmation of parameter availability. Detailed raw data extraction will follow in the later validation phase.
• Transition to parameterisation + scenario development phase	🕒 In Progress	Progress in this step depends on validating KPIs and data availability with the partners.
Need for structured coordination identified		We propose a fixed coordination structure (bi-weekly WP3.3 meetings, monthly WP3–WP4 & WP3–WP2 syncs, and quarterly review packages).

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Current Status

1. KPIs have been analysed.
2. Value chain flows have been created for both use cases.
3. A preliminary simulation model has been developed.

Completed Work

For each use case, KPI sets were identified based on the provided Use Case Descriptions.

Not all of these KPIs fall within the scope of the Value Chain Digital Twin; therefore, the model will prioritise the following core KPIs:

Lead-time (total and inter-stage)

On-Time Delivery (OTD)

Inventory levels (raw, semi-finished, finished)

Overstock / stockout risk

Transport-related delays

Waiting time and bottleneck identification

Flow stability

Chain-level resilience (supplier delays, capacity loss, variability)

Production planning efficiency (equipment assignment, sequencing rules)

Forecasting accuracy and its effect on inventory and throughput

Flow-level energy consumption (frequency of transitions, thermal cycle impact)

Supplier risk propagation (single- vs multi-sourcing resilience)

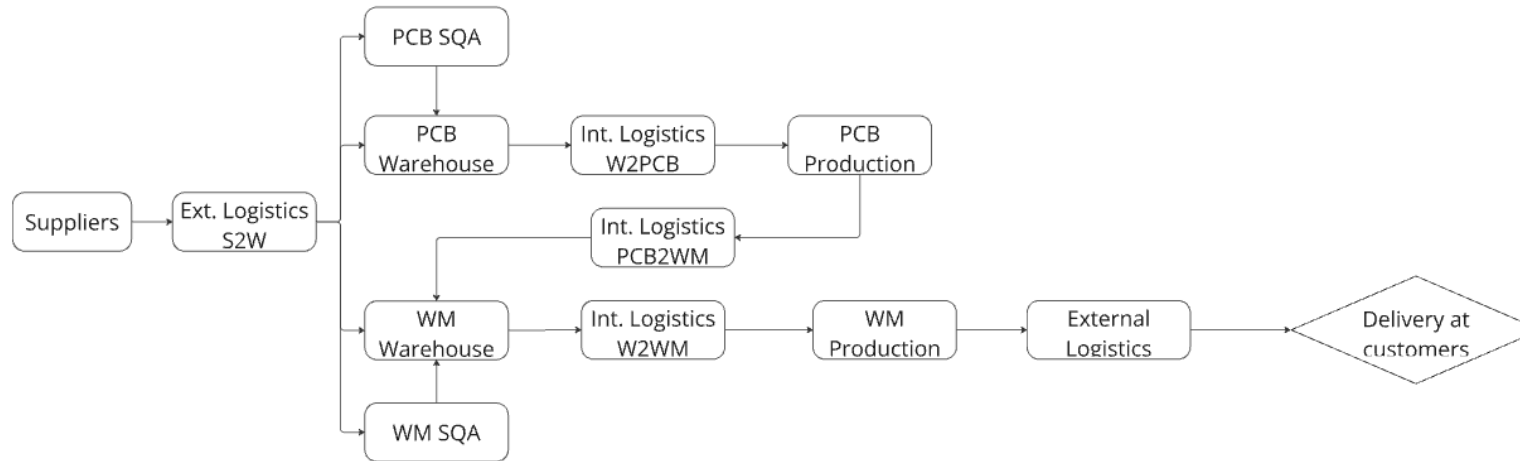
Inventory value behaviour (holding-cost trends driven by flow patterns)

Please let us know if these KPIs meet your expectations.

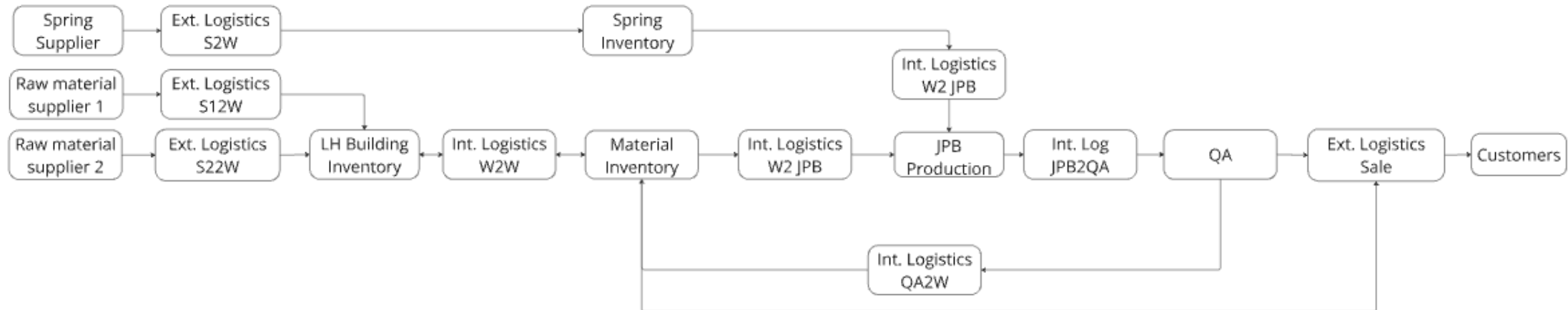
Additional KPIs can be incorporated in later phases if needed.

Completed Work

KAM



JPB



Completed Work

Characterization	Input	Output	Targeted KPI
Supplier	Supplier order (Product)		- Resources sourcing - Stock optimization
	Quantity of each product		- Resources sourcing - Stock optimization
	Expected delivery time	Actual delivery time	- DeliveryTime - Resources sourcing - Stock optimization - Actual cycle time compared to optimal cycle time
	Ordering cost		- Resources sourcing - Cost & Resilience
		Unit price of each product	- Resources sourcing - Stock optimization - Cost & Resilience
		Order fill rate	- Resources sourcing - Stock optimization - Cost & Resilience

Outbound Logistics			
	Departure time		- DeliveryTime - Resources sourcing - Stock optimization
	Transport duration		- DeliveryTime - Resources sourcing - Stock optimization
	Products		- Resources sourcing - Stock optimization
	Expected delivery time	Actual delivery time	- DeliveryTime - Resources sourcing - Stock optimization - Actual cycle time compared to optimal cycle time
	List of products		- Resources sourcing - Stock optimization
	Defective product rate		- Resources sourcing - Stock optimization
	Purchasing Lead Time		

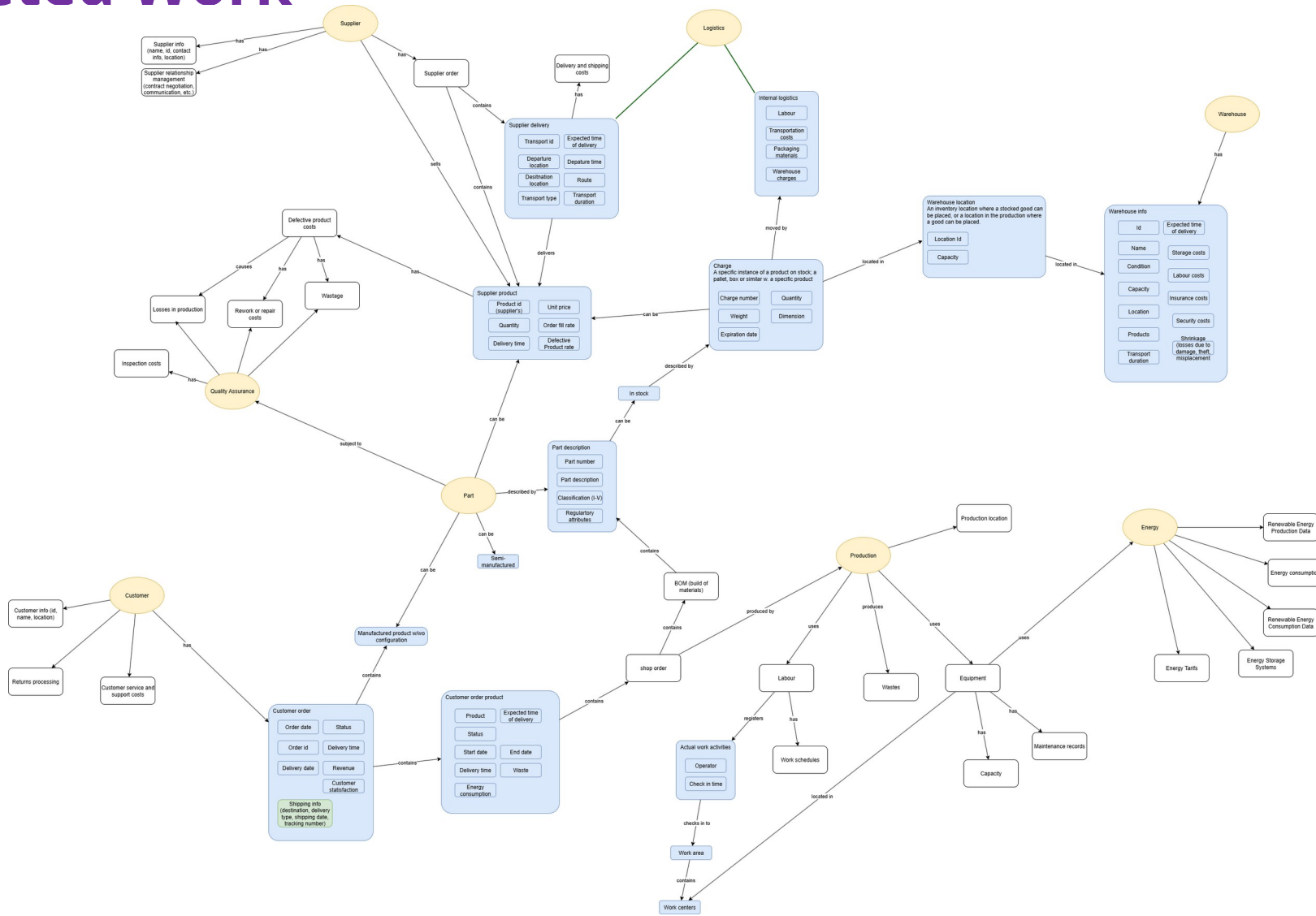
Warehouse	Quantity of each product		
	Holding cost		
	Max capacity		
	Stock out cost		
		Idle	- Resources sourcing - Cost & Resilience
		Output Quantity	- Resources sourcing - Stock optimization
	Weight, Dimension		- Resources sourcing - Stock optimization

Inbound Logistics			
	Departure time		- DeliveryTime - Resources sourcing - Stock optimization
	Transport duration		- DeliveryTime - Resources sourcing - Stock optimization
	Products		- Resources sourcing - Stock optimization
	Expected delivery time	Actual delivery time	- DeliveryTime - Resources sourcing - Stock optimization - Actual cycle time compared to optimal cycle time
		Defective product rate	- Resources sourcing - Stock optimization - Cost & Resilience

Production			
	Customer order		Resources sourcing
	List of products		- Resources sourcing - Stock optimization
	Quantity of products		- Resources sourcing - Stock optimization
	Expected production time	Actual Production time	- DeliveryTime - Resources sourcing - Stock optimization - Actual cycle time compared to optimal cycle time
	Labour		
	Machinery		
	Energy		
	How many people works		- Resources sourcing - Cost & Resilience
	how many machine works		- Resources sourcing - Cost & Resilience
	Cost		- Resources sourcing - Cost & Resilience
	Waste		- Resources sourcing - Cost & Resilience - Energy consumption & Sustainability
	Energy consumption		- Resources sourcing - Cost & Resilience - Energy consumption & Sustainability
	Environmental impact		- Resources sourcing - Energy consumption & Sustainability

QA			
	Customer order	Order fill rate	- Resources sourcing - Stock optimization
	Products	Defected product rate	- Resources sourcing - Stock optimization
		Losses	- Resources sourcing - Cost & Resilience - Energy consumption & Sustainability - The Operations Management perspective.
		Time losses	- Resources sourcing - Cost & Resilience - Energy consumption & Sustainability - The Operations Management perspective.
		Speed losses	- Resources sourcing - Cost & Resilience - Energy consumption & Sustainability - The Operations Management perspective.
		Equipment losses	- Resources sourcing - Cost & Resilience - Energy consumption & Sustainability - The Operations Management perspective.
		Operation losses	- Resources sourcing - Cost & Resilience

Completed Work



Current Status and ToDo's

Home

Stock Based Simulation

FIFO Simulation

Comparison Analysis

Simulation Settings

Simulation Time (days)

136

1190

Initial Stock

53

Safety Stock Level

20

Manufacturing Lead Time (days)

5,00

Shipping Time (days)

3,00

Supply Lead Time (days)

2,00

OTIF Threshold (days)

9,00

Stock-based Supply Chain Simulation

Run Stock-based Simulation

Running simulation... Please wait.

Simulation completed.

Daily Simulation Data

Day	Orders	Produced	Shipments	Stock
0	1	0	0	53
1	2	0	0	53
2	3	0	0	53
3	4	0	0	53
4	5	85	0	53
5	6	0	0	53
6	7	0	0	53
7	8	94	0	53
8	9	0	0	53
9	10	19	0	53

Simulation Metrics

Total Demands: 7

OTIF (<= 9.0 days): 42.86%

Average Delivery Time: 12.71 days

Total Production: 216

Final Stock: 55

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Remaining Tasks & Next Steps

[illegible]

Remaining Tasks & Next Steps

Work	Tasks/Activities	Start	END
Parameterisation & KPI Alignment	•KPI extraction and validation with partners •Literature review on Value Chain Digital Twins, DES, multi-layer models, KPI frameworks, resilience modelling •Collection of VC-DT requirements (functional + non-functional) •Definition of value-chain parameters for all 6 layers •Development of a VC-DT UML diagram (class, sequence, activity diagrams as applicable) •Initial software architecture definition for the VC-DT •Scenario library v1 creation •Initial data requirements consolidation	W1	W12
Simulation Model Development	• Development of full VC-DT DES engine (SimPy) • Modelling of delays, flows, queue behaviour, inventory dynamics • Implementation of modular layer-based architecture • Integration of stochastic parameters (fixed, KDE-based, or fitted distributions)	W9	W28
Extended Data Exploration & Pre-Validation	• Parallel early exploration of raw data provided by partners • Identification of available fields vs. required model inputs • Data cleaning trials (format inconsistencies, missing values, timestamp alignment) • Mapping raw dataset → parameter table variables • Detecting data limitations influencing KPI calculation • Pre-validation runs comparing expected vs. simulated behaviour • Recording uncertainties requiring partner clarification	W5	W32

Remaining Tasks & Next Steps

Work	Tasks/Activities	Start	END
WP2, WP4 and Task 3.2 Alignment	•Continuous alignment with WP2, WP3.2 and WP4 to ensure VC-DT requirements reflect upstream work •WP2 → ontology, DKG structures, data accessibility •WP3.2 → manufacturing-level DT behaviours, constraints •WP4 → DSS input format, UI needs, scenario types •Collecting incoming outputs and design decisions from WP2/WP3.2/WP4 and updating VC-DT requirements accordingly •Extraction and cleaning of real factory data (KAM & JPB) •Mapping raw data → VC-DT simulation parameters •Validation: simulation vs. real system behaviour (lead-time, stock, delays, disruptions) •Sensitivity & error analysis •Iterative refinement of model parameters and architecture based on cross-WP inputs	W25	W40
Finalisation, DSS & UI Integration	• Generation of structured outputs for WP4.3 learning agents • JSON / API output schema design for DSS • Integration planning with WP4.2 (UI for scenario testing) • Final scenario library	W37	W56
Testing	•Testing	W40	W56
Continuous Reporting & Documentation (W1–W56)	•Weekly documentation of modelling decisions •Capturing WP2/WP3.2/WP4 alignment notes •Updating VC-DT assumptions, diagrams, requirements •Keeping a running draft of D3.2 sections •Logging data exploration findings as they occur •Integrating simulation results continuously into the report	W1	W56

Remaining Tasks & Next Steps

The preliminary model will be expanded to build a full Value Chain Digital Twin representing the entire system.

At this stage:

- The KPIs provided by the partners need to be confirmed.
- We should clarify whether there are any additional priorities or KPI expectations.
- We need to discuss how the data will be reviewed and validated during the upcoming phases.

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Coordination Plan with Other Partners

WP2

Open Points – WP2 Integration & Data Validation Needs

Bullets:

- Expected timeline for completing the ontology and establishing the entity links
- When the DKG infrastructure and query mechanism will be operational
- How WP3.3 should proceed until ontology + DKG integration is ready
- Recommended approach for working with manufacturer partners during this interim phase
- Support needed from WP2 for extracting and validating real factory data

Coordination Plan with Other

Partners WP3.2 and WP3.2 Integration

How should we plan the integration between Task 3.2 and Task 3.3 once the VC-DT model is mature?

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Coordination Plan with Other

Partners

WP4.3 MO-DDSS Module

- Does the current WP3.3 work adequately support your needs?
- How should the Value Chain DT and simulation outputs best feed your activities?
- What type of inputs/outputs or coordination would be most useful for you?

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Open Discussion

- Open discussion points?
-
- Remarks?
-
- Challenges?

Reminder

- Next meeting

THANK YOU FOR
YOUR ATTENTION



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